Practical 11 – Static Routing and RIP

AIM

To set up static routes and RIP routing between different networks and check their operation in Cisco Packet Tracer.

REQUIREMENTS

Cisco Packet Tracer, routers, PCs, and suitable Ethernet or serial cables.

PROCEDURE

1. Add three routers and the necessary PCs to the Packet Tracer workspace.
2. Link the devices with the appropriate network cables.
3. Configure IP addresses and subnet masks on router interfaces and PCs, and set each PC's default gateway.
4. Activate router interfaces with `no shutdown`. Set a clock rate on a DCE serial interface if needed.
5. Add static routes with `ip route`; configure an alternate route when required.
6. Check static entries with `show ip route static`.
7. Configure RIP using `router rip`, select version 2, and advertise the connected networks.
8. Use `ping` and `tracert` to test communication.

SAMPLE STATIC ROUTE

```
ip route [destination-network] [subnet-mask] [next-hop-IP]
```

SAMPLE RIP CONFIGURATION

```
router rip
version 2
network [network-address]
no auto-summary
```

INPUT

Router interface addresses, destination networks, subnet masks, next-hop IP addresses, and PC gateway settings.

OUTPUT

The routing table shows static entries or routes learned through RIP. Ping and tracert results indicate whether the PCs can communicate.

OBSERVATION

1. Three routers and the PCs were connected, and IP details were assigned to the interfaces and end devices.
2. Router interfaces were enabled; a clock rate was entered on the DCE serial link where required.
3. Static routes and any alternate paths were added and checked in the routing table.
4. RIP version 2 was configured and the relevant networks were advertised.
5. Ping and tracert were used to verify the route and end-to-end connectivity.

RESULT

Static routing and RIP were configured and verified successfully in Cisco Packet Tracer.